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Abstract: Temporal interference neurostimulation (TI) offers a novel opportunity to noninvasively stimulate deep brain regions without modulating overlying cortex. This key advantage over other noninvasive brain stimulation methods (e.g., transcranial magnetic stimulation, transcranial direct current stimulation) presents a unique tool for causal manipulation of deep brain structures within large-scale networks. However, the effects of TI on whole-brain functional connectivity remain largely unexplored. To address this gap, we conducted a psychophysiological interaction (PPI) analysis using functional MRI data from 20 human participants who received active TI stimulation of the mid-orbitofrontal cortex (OFC) at a 20 Hz beat frequency as well as sham stimulation delivered in four alternating 8-minute blocks. Results revealed a significant functional connectivity effect of TI, with a significant PPI interaction term between the OFC and the left parahippocampal gyrus ($p < 0.001$) during stimulation, suggesting network-level modulation involving regions linked to affective and contextual processing. This is the first study to demonstrate changes in functional connectivity induced by TI stimulation of the orbitofrontal cortex. These findings support the potential of TI as a precise and scalable method for investigating and manipulating causal dynamics within distributed brain networks, offering new avenues for connectivity research in both basic and translational neuroscience.

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Poster

PSTR147: Stimulation Methods

Location: SDCC Halls B-H

Time: Sunday, November 16, 2025, 1:00 PM - 5:00 PM

Program #/Poster #: PSTR147.14/RR11

Topic: J.08. Methods to Modulate Neural Activity

Support: BMBF AI.D / 16ME0541

Title: Exploring neuronal response dynamics to intracortical electrical microstimulation in human cognitive association cortex

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Abstract: Intracortical electrical microstimulation (ICMS) can modulate neural activity and reveal causal contributions of single neurons and neuronal populations to behavior. ICMS has been successfully used to influence sensory perception, motor actions, and decision-making in

animals and humans. However, its effects on single-neuron spiking remain poorly understood, particularly in the human frontoparietal association cortex—a region critical for complex cognitive functions and uniquely expanded in humans. Here, we applied ICMS in a subject with stroke-induced language impairments (aphasia) through planar microelectrode arrays (64 electrodes each) chronically implanted in the inferior and middle frontal gyri (IFG, MFG) of the right hemisphere. Implantation targeted regions homotopic to the left language network to explore the potential of targeted neuromodulation with cellular resolution. To balance resting-state and task engagement, we used a passive fixation task minimizing behavioral confounds while maintaining stable attention. After 1000 ms of stable fixation, ICMS was delivered as a single pulse or a train lasting 500-800 ms, followed by a 3200-3500 ms post-stimulation period. Stimulation parameters included five current amplitudes (1, 5, 10, 20, 40 µA) and five frequencies (single pulse, 4, 8, 20, 76 Hz). Sham trials (no stimulation) were included, and all conditions were pseudorandomized. Stimulation artifacts were removed using local polynomial fitting, allowing precise analysis of post-stimulation spiking activity. In a representative session alternating stimulation between two MFG electrodes, 70 multi-units were detected across 48 electrodes. Stimulation responses were quantified relative to baseline (fixation epoch) firing rates. Stimulation trains combining high frequency (76 Hz) and high amplitude (20, 40 µA) elicited widespread inhibition on 47 electrodes within 100 ms of stimulation offset. For single pulses and trains with lower frequency or amplitude, modular responses appeared throughout the post-stimulation epoch. Across multiple months of concurrent stimulation and recording, signal and noise characteristics remained stable as measured by RMS and microelectrode impedances. Cumulative charge did not correlate with impedance or RMS changes. No adverse clinical events occurred. These observations support the safety of the stimulation protocol and the stability of long-term intracortical recordings. We believe our findings will contribute to laying groundwork for future investigations into the causal roles of defined populations of human neurons to higher brain functions, particularly language.

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Poster

PSTR147: Stimulation Methods

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Program #/Poster #: PSTR147.15/RR12

Topic: J.08. Methods to Modulate Neural Activity

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Title: Does electrical stimulation modulate the magnitude and generalization of neurophysiologic effects of prism adaptation?